

Double Radiation Cancellation at an Acoustic Rayleigh Threshold

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Bound states in the continuum require complete cancellation of radiation from an eigenmode. At a diffraction threshold, however, vanishing normal flux does not imply confinement, because a finite grazing field remains nondecaying and non-square-integrable. We establish a strict Rayleigh-threshold bound state in a periodic underwater acoustic surface. The homogeneous state occurs exactly when a first diffraction order opens, while both its threshold amplitude and the amplitude of the remaining propagating order vanish. A full outgoing-wave operator, pole continuation, and exterior-norm saturation distinguish the state from a Rayleigh anomaly and a high- Q resonance. Two groove-supported source families independently cancel threshold and finite-flux radiation. These results identify channel-opening BICs as a threshold subclass governed by amplitude cancellation rather than zero flux alone.

A bound state in the continuum (BIC) is a spatially localized eigenstate at a real frequency embedded in a radiation continuum. Its defining property is the complete disappearance of all outgoing components of a *homogeneous* mode, rather than a narrow feature in a driven spectrum [1–3]. Symmetry incompatibility, Friedrich–Wintgen interference, Fabry–Pérot trapping, and topological radiation zeros provide established routes to this decoupling [4–6]. These mechanisms are normally formulated where the set of asymptotic radiation channels remains fixed.

A diffraction threshold is qualitatively different. When a Floquet order reaches grazing propagation, its normal wavenumber and normal power flux vanish, yet a finite pressure amplitude remains independent of the surface-normal coordinate and is therefore not square-integrable. Thus a Rayleigh anomaly does not become a BIC merely because the grazing order carries no normal power. A strict threshold state must cancel the grazing amplitude itself together with every other propagating amplitude. This distinction connects the square-root singularity of threshold scattering [7, 8] to the localization condition for an embedded eigenstate.

Recent electromagnetic work describes a resonant pole reaching a Rayleigh branch point [9], while acoustic BICs have been realized through resonator interference, symmetry, and underwater Fabry–Pérot trapping [10, 11]. What remains unresolved in scalar acoustic diffraction is whether a homogeneous mode can be localized exactly as a new channel opens while another diffraction order is already propagating. We establish such a state in a periodic underwater acoustic surface and test the threshold coefficient directly.

The state is formed by channel-resolved destructive interference. Two groove-supported source families acquire independent internal phases and close separate radiation sums for the grazing and finite-flux channels. A null of the complete outgoing-wave operator, continuation of the associated pole, and saturation of the exterior norm distinguish this state from both a generic Rayleigh anomaly and a finite-linewidth resonance. The result identifies

channel-opening BICs as a threshold subclass for which field-amplitude cancellation, rather than zero flux alone, restores localization.

Consider a sound-hard surface of period a occupying $y \leq 0$, with water in $y > 0$ and time dependence $e^{+i\omega t}$. For incidence toward the surface and outgoing reflection,

$$p_{\text{inc}} = e^{-ik_{x,0}x + ik_{y,0}y}, \quad p_{\text{sc}} = \sum_{n \in \mathbb{Z}} A_n^{\text{dr}} e^{-ik_{x,n}x - ik_{y,n}y}, \quad (1)$$

$$k_{x,n} = k_0 \sin \theta + \frac{2\pi n}{a}, \quad k_{y,n} = \sqrt{k_0^2 - k_{x,n}^2}. \quad (2)$$

The outgoing branch has $\Re k_{y,n} \geq 0$ and $\Im k_{y,n} \leq 0$. We use

$$\Omega = \frac{af}{c}, \quad \kappa = \frac{ak_0 \sin \theta}{2\pi} = \Omega \sin \theta, \quad |\kappa + n| = \Omega \quad (3)$$

for a Rayleigh threshold of order n .

The two-groove cell is shown in Fig. 1; its complete dimensional specification is given in the Supplemental Material. At $f = 200$ kHz and $c = 1500$ m/s,

$$\begin{aligned} \kappa_{\text{BIC}} &= 0.083848696553, & \Omega_{\text{BIC}} &= 0.916151303447, \\ \theta_{\text{BIC}} &= \pm 5.251216^\circ. \end{aligned} \quad (4)$$

For $+\theta$, $n = 0$ is propagating, $n = -1$ is at threshold because $\kappa + \Omega = 1$, and $n = +1$ is evanescent. Reversing the Bloch number exchanges $n = -1$ and $n = +1$.

Inside groove ℓ , a bottom-rigid cosine expansion is retained. Projecting pressure and normal-velocity continuity without eliminating the groove coefficients gives the pole-free homogeneous system

$$F(\omega, \kappa; \mathbf{g}) \begin{bmatrix} \mathbf{A}^{\text{hom}} \\ \mathbf{C} \end{bmatrix} = 0, \quad \mathbf{g} = (d_1, d_2, w_1, w_2, g). \quad (5)$$

The full block matrix, scaling, and driven reduction are given in the Supplemental Material. This separation is essential: A_n^{dr} contains the forced background response, whereas A_n^{hom} belongs to an outgoing eigenmode.

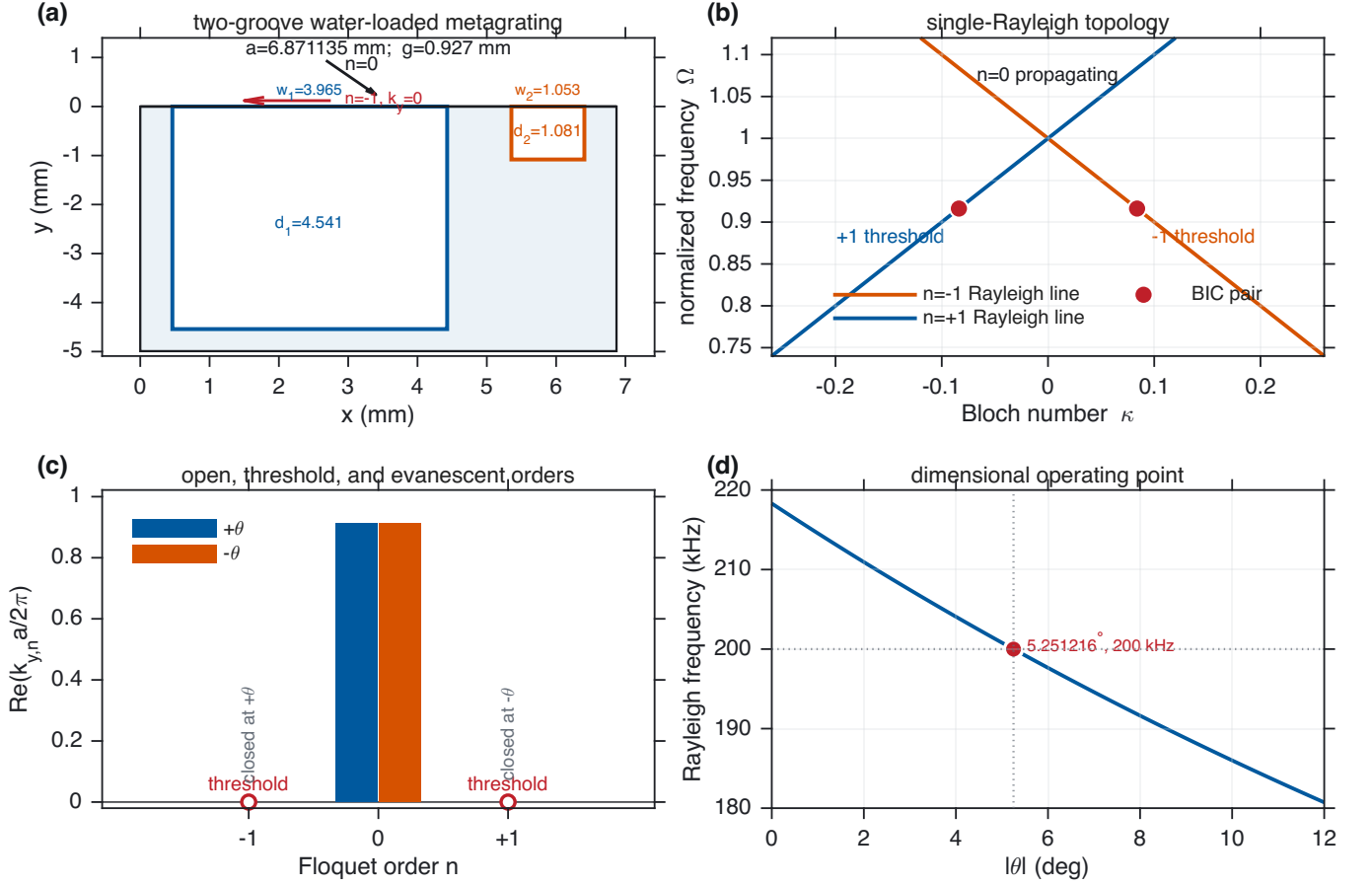


FIG. 1. Acoustic realization and channel topology. (a) Two-groove cell; dimensions are in millimeters. (b) The two first-order Rayleigh lines and the reciprocal BIC pair. The zeroth order is propagating at both points. (c) Normal wavenumbers for the three displayed orders: for $+\theta$ ($-\theta$), $n = -1$ ($n = +1$) is grazing and the opposite first order is evanescent. (d) The positive-angle point on the dimensional $n = -1$ Rayleigh curve.

For a single homogeneous exterior order, define the height-integrated pressure norm $\mathcal{N}_n(H)$ by

$$\mathcal{N}_n(H) = \int_0^H |p_n^{\text{hom}}|^2 dy. \quad (6)$$

It evaluates to

$$\mathcal{N}_n(H) = \begin{cases} H|A_n^{\text{hom}}|^2, & k_{y,n} = 0, \\ \frac{|A_n^{\text{hom}}|^2}{2\alpha_n}(1 - e^{-2\alpha_n H}), & k_{y,n} = -i\alpha_n. \end{cases} \quad (7)$$

Thus the positive-angle state is strict only if

$$A_{-1}^{\text{hom}} = 0, \quad A_0^{\text{hom}} = 0. \quad (8)$$

We first impose Eq. (8) by removing those two amplitude columns from the unknown vector, row- and column-scale the remaining operator, and reconstruct its smallest right singular vector. At $(N, K) = (313, 39)$, where N is the number of exterior orders and K is the cosine-mode count

per groove,

$$\begin{aligned} \sigma_{\min}/\sigma_{\max} &= 1.2603643 \times 10^{-15}, \\ r_{\text{raw}} &= 1.3765668 \times 10^{-13}. \end{aligned} \quad (9)$$

This constrained null is not the sole evidence. Evaluating the complete complex-frequency operator at the same final geometry, with the selected $k_{y,-1}$ set exactly to zero, gives $\sigma_{\min}/\sigma_{\max} = 7.7151 \times 10^{-16}$ and an unconstrained null whose open and threshold amplitudes vanish at machine precision.

The null persists under modal refinement, and a same-geometry complex-pole continuation remains on one branch while approaching the real Rayleigh endpoint [Fig. 2]. These tests support a converging embedded eigenstate rather than a real-axis minimum generated by the threshold singularity. The truncation sequence, mode overlaps, quality factors, and finite-model scaling fit are reported in the Supplemental Material.

Figure 3 provides a spatial test. The strict mode contains only evanescent exterior orders, and its exterior norm saturates within a few periods. Restoring any

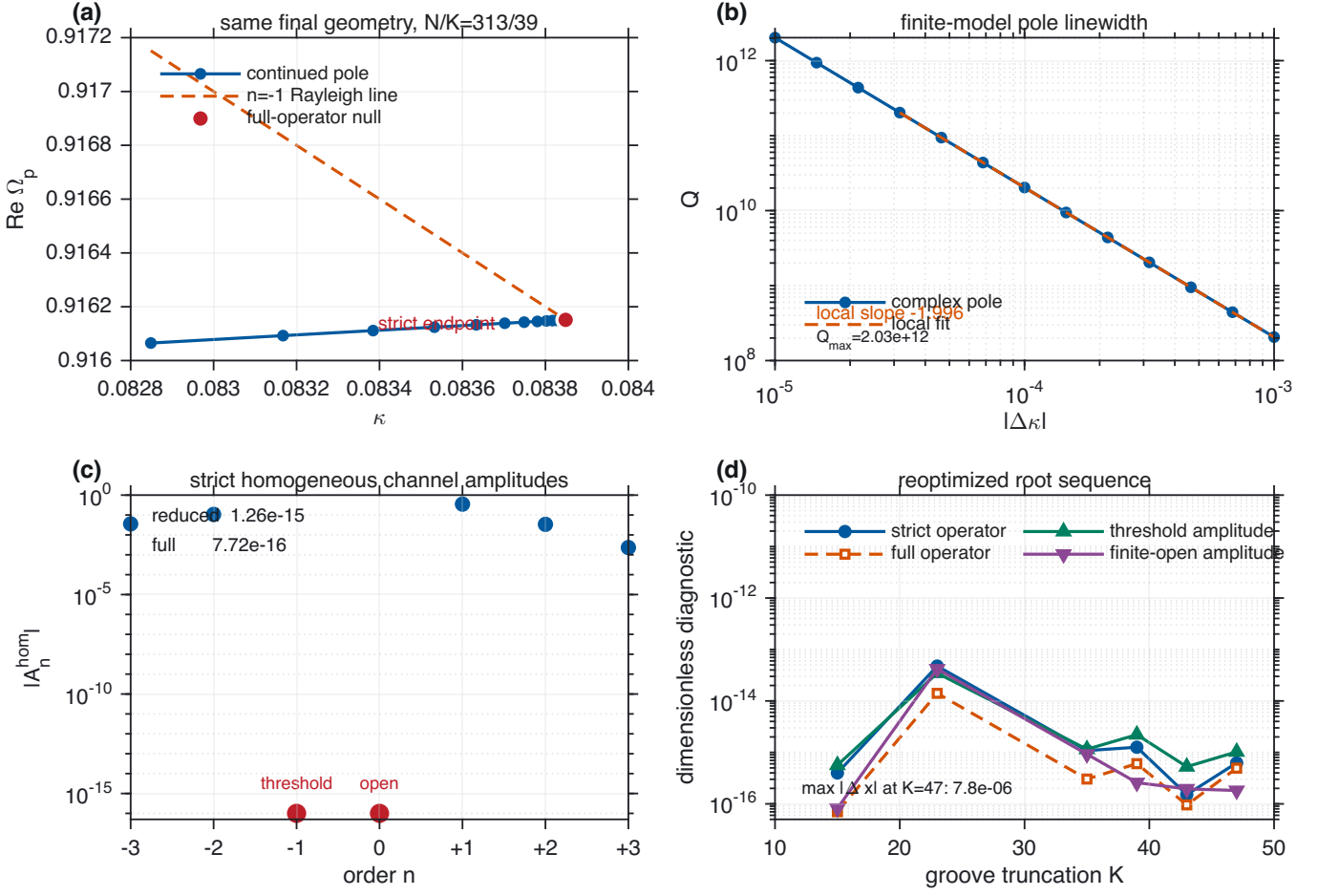


FIG. 2. Strict-null evidence. (a) Final-geometry pole trajectory and the $n = -1$ Rayleigh line. The red endpoint is also a null of the full (313, 39) operator. (b) Pole quality factor and local fit; the exponent is a resolved finite-model trend. (c) Final homogeneous Floquet amplitudes. The finite-flux $n = 0$ and threshold $n = -1$ coefficients vanish, while the displayed remaining orders are evanescent. (d) Reoptimized root sequence: reduced and full singular-value ratios and the independently extracted threshold/open amplitudes remain at machine precision through (377, 47).

nonzero grazing amplitude produces the linear term in Eq. (7). Near-BIC and ordinary-resonance real-axis surrogates are shown only as discriminating controls; their reconstruction and channel errors are detailed in the Supplemental Material.

The two zero conditions can be resolved into aperture-source phasors,

$$A_m^{\text{hom}} = \sum_{\ell=1}^2 \sum_q \mathcal{R}_{m,\ell q} C_{\ell q} = 0, \quad m \in \{-1, 0\}. \quad (10)$$

The two grooves close the $n = -1$ and $n = 0$ phasor polygons separately, with normalized closure errors 1.6×10^{-15} and 1.1×10^{-15} . This motivates, but does not prove universally, the need for two independent phase controls. A global search over a *single large rectangular* groove, retaining transverse modes and varying $0.04 \leq \kappa \leq 0.25$, $0.04 \leq d/a \leq 2$, and $0.50 \leq w/a \leq 0.95$, reaches only $\sigma_{\min}/\sigma_{\max} = 0.124588$ at (221, 44). The comparison is therefore a bounded ablation within this

declared shape family, not a no-go theorem for arbitrary one-cavity structures.

Fabrication perturbations, dimensional similarity, sound-speed detuning, and driven scattering are reported in the Supplemental Material. They delimit the ideal eigenstate result but are not used to certify it. In particular, the driven response is not interpreted as a converged one-way device, and a separate high-efficiency off-Rayleigh point is an ordinary anomalous reflector rather than a BIC signature.

We have established a strict acoustic bound state at a Rayleigh channel opening. Localization requires cancellation of both the zero-flux threshold amplitude and the coexisting finite-flux radiation amplitude. The two-groove realization exposes these conditions as separate interference constraints, identifying channel-opening BICs as a threshold subclass of embedded eigenstates. The result applies to an infinite, rigid, lossless acoustic surface; finite-size leakage, dissipation, elastic compliance,

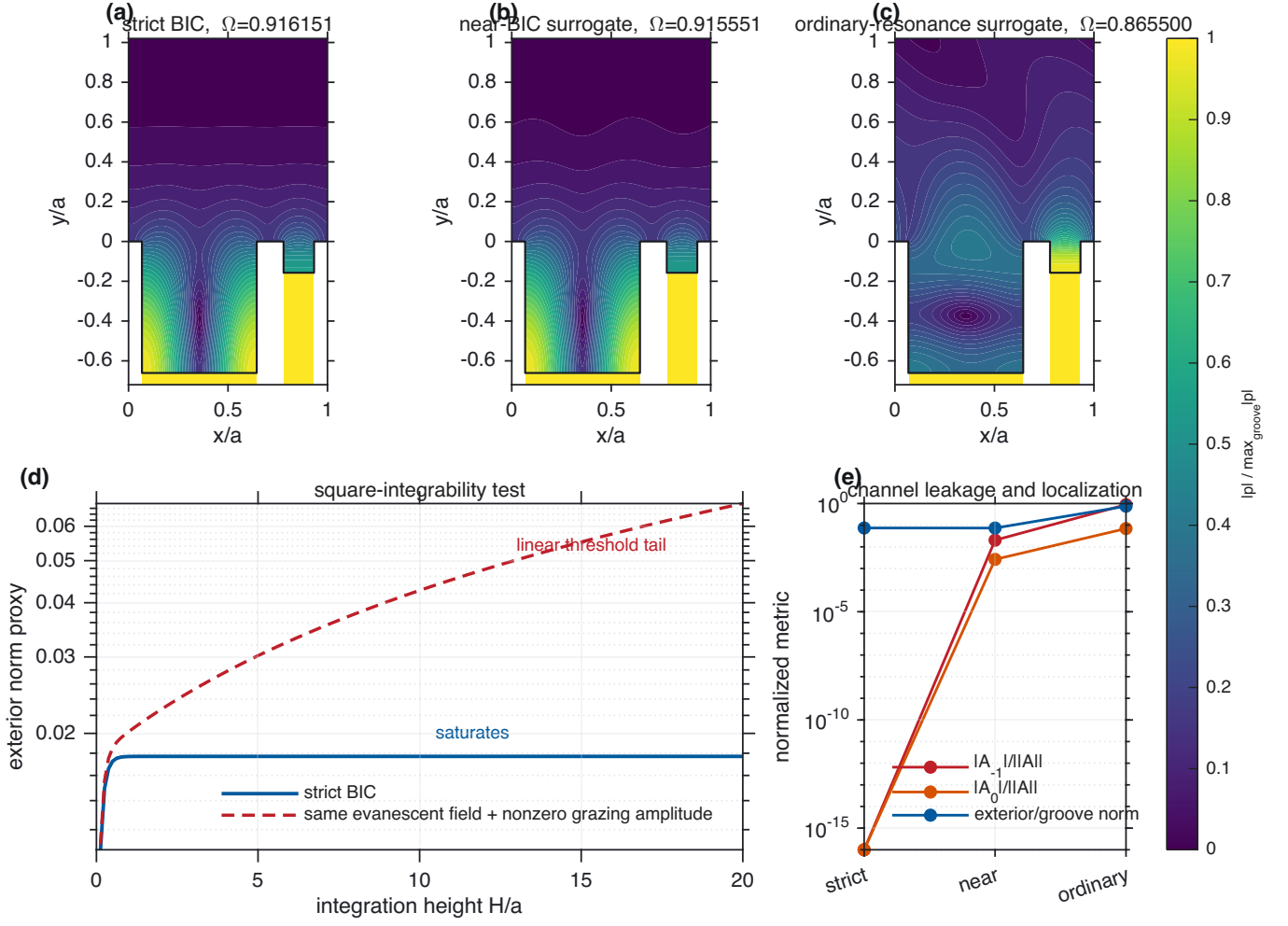


FIG. 3. Homogeneous pressure fields and localization. (a) Strict BIC. (b) Near-BIC real-axis surrogate at $\Omega = 0.9155513$. (c) Ordinary-resonance real-axis surrogate at $\Omega = 0.8655$. All maps show $|p| / \max_{\text{groove}} |p|$ on the same scale; white regions are rigid material. (d) Exterior norm versus integration height for the strict mode and for the same evanescent field with a nonzero grazing component added. (e) Threshold amplitude, finite-open amplitude, and exterior-to-groove norm proxies. Panels (b,c) are diagnostic surrogates, not exact quasinormal-mode fields.

and experimental realization remain outside the present claim.

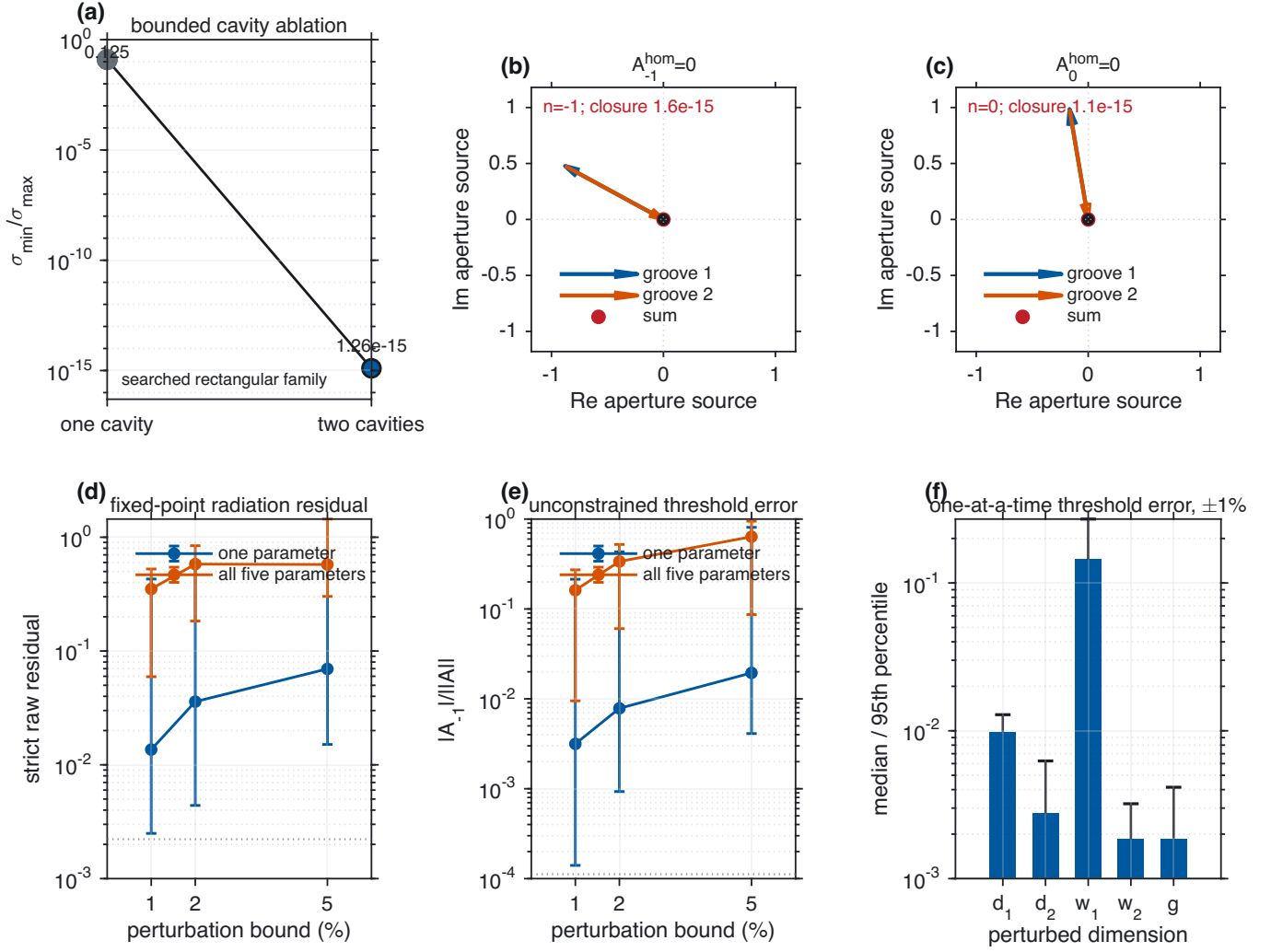


FIG. 4. Cavity degrees of freedom and fixed-point sensitivity. (a) Best single-large-cavity result in the searched rectangular family versus the two-groove root. (b,c) Separate aperture-source phasor closures for the grazing $n = -1$ and propagating $n = 0$ constraints. (d,e) Medians and 5–95% intervals of the strict raw residual and the unconstrained threshold-amplitude error for one-parameter and joint perturbations. The dotted line is the unperturbed (185, 23) numerical floor. (f) One-at-a-time $\pm 1\%$ sensitivity; bars are medians and whiskers are 95th percentiles.

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- [1] C. W. Hsu, B. Zhen, A. D. Stone, J. D. Joannopoulos, and M. Soljačić, Bound states in the continuum, *Nature Reviews Materials* **1**, 16048 (2016).
 - [2] K. Koshelev, Z. F. Sadrieva, A. A. Shcherbakov, Y. S. Kivshar, and A. A. Bogdanov, Bound states in the continuum in photonic structures, *Physics-Uspekhi* **66**, 494 (2023).
 - [3] S. I. Azzam and A. V. Kildishev, Photonic bound states in the continuum: From basics to applications, *Advanced Optical Materials* **9**, 2001469 (2021).
 - [4] Y. Yang, C. Peng, Y. Liang, Z. Li, and S. Noda, Analytical perspective for bound states in the continuum in photonic crystal slabs, *Physical Review Letters* **113**, 037401 (2014).
 - [5] B. Zhen, C. W. Hsu, L. Lu, A. D. Stone, and M. Soljačić, Topological nature of optical bound states in the continuum, *Physical Review Letters* **113**, 257401 (2014).
 - [6] C. W. Hsu, B. Zhen, J. Lee, S.-L. Chua, S. G. Johnson, J. D. Joannopoulos, and M. Soljačić, Observation of trapped light within the radiation continuum, *Nature* **499**, 188 (2013).
 - [7] L. Rayleigh, On the dynamical theory of gratings, *Proceedings of the Royal Society of London A* **79**, 399 (1907).
 - [8] C. C. Wojcik, H. Wang, M. Orenstein, and S. Fan, Universal behavior of the scattering matrix near thresholds in photonics, *Physical Review Letters* **127**, 277401 (2021).
 - [9] I. Karavaev, M. Song, and A. Bogdanov, Rayleigh bound states in the continuum (2026), arXiv:2607.07228v1, arXiv:2607.07228 [physics.optics].
 - [10] M. Farhat, M. Achaoui, J. A. I. Martínez, M. Addouche, Y. Wu, and A. Khelif, Observation of ultra-high- q resonators in the ultrasound via bound states in the continuum, *Advanced Science* **11**, 2402917 (2024).
 - [11] Y. Yang, H. Jia, Y. Yang, P. Zhou, and J. Yang, Observation of merged fabry–perot bound states in the continuum within an underwater phononic crystal, *Physical Review B* **110**, 184108 (2024).